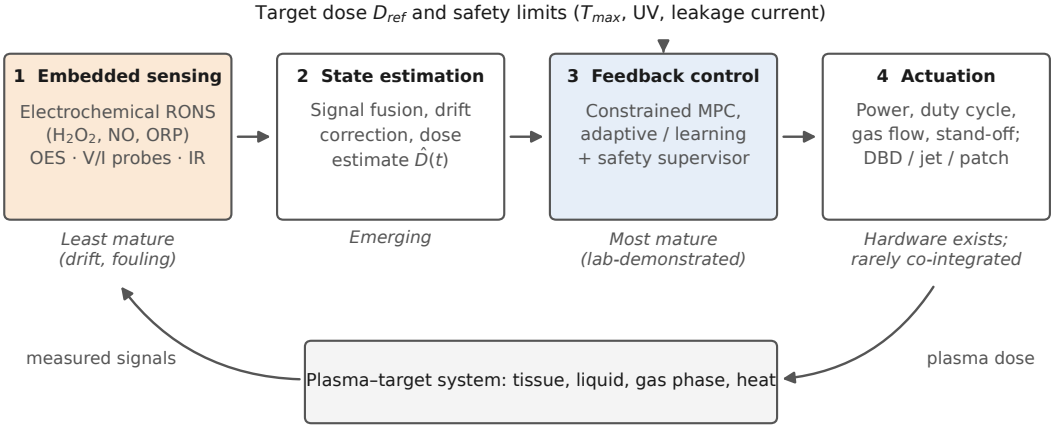
Target dose D_{ref} and safety limits (T_{max}, UV, leakage current)
1 Embedded sensing
Electrochemical RONS (H_2O_2, NO, ORP)
OES · V/I probes · IR
2 State estimation
Signal fusion, drift correction, dose estimate $\hat{D}(t)$
3 Feedback control
Constrained MPC, adaptive / learning + safety supervisor
4 Actuation
Power, duty cycle, gas flow, stand-off; DBD / jet / patch
Least mature (drift, fouling)
Emerging
Most mature (lab-demonstrated)
Hardware exists; rarely co-integrated
measured signals
Plasma–target system: tissue, liquid, gas phase, heat
plasma dose